MY PRINCIPLES ON AI IN MATHEMATICS

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I have signed the Leiden Declaration on Artificial Intelligence and Mathematics and A Severe Misalignment of AI in Mathematics.

Some mathematicians have explicitly stated that they do not use LLMs in their mathematical research; see, for example, Vladimir Lazić's position. Their voices deserve serious attention.

I use LLMs and find them genuinely valuable. At this stage, I set out the following principles as my personal position; I do not expect other mathematicians to make the same choices.

1. **What is retrieved from an LLM is not research; its credit belongs to the human thought from which it derives.**

   An LLM is trained on the accumulated work of generations of mathematicians, and it is available to anyone who asks. What it generates should therefore be regarded as public knowledge, much like the literature: its insight derives from past human thought, and anyone could retrieve it. Retrieving it, however skillfully, is not an original contribution.

2. **The main contribution a paper claims should come from the author's own thinking.**

   I use LLMs as I would consult the literature: I ask them questions and learn from their answers. But the main ideas of my papers are developed by myself, not in discussion with an LLM. An idea that emerges from such a discussion cannot be cleanly attributed to the researcher, and since the same model is available to everyone, it does not reflect the insight of whoever happened to retrieve it first.

3. **Distinguishing the contributions of LLMs from one's own matters, both for one's own development and for one's academic reputation.**

   When what comes from an LLM is acknowledged as such, what remains is genuinely one's own. This allows the researcher to see their own development honestly, and it allows readers to trust what a paper claims as its contribution, which is the basis of a researcher's reputation.

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